STUDY.md

Solana and SVM Study Topics

Overview

This document outlines the study topics for mastering Solana and SVM (Solana Virtual Machine), with direct comparisons to EVM (Ethereum Virtual Machine) concepts where applicable.

Each topic includes key learning objectives and implementation considerations for this project.

Core Concepts

1. Accounts and Programs

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- **Solana Equivalent**: Accounts (data storage) and Programs (smart contracts)
- **EVM Comparison**: Similar to contracts and storage, but accounts hold both code and data

Key Topics - Accounts & Programs

- Account types: System accounts, program accounts, data accounts
- Rent exemption and account lifecycle
- Program Derived Addresses (PDAs) for deterministic addressing

Implementation: Account management API endpoints

2. Transactions and Instructions

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- **Solana Equivalent**: Transactions containing Instructions
- **EVM Comparison**: Transactions calling contract functions

Key Topics - Transactions & Instructions

- Transaction structure and serialization
- Instruction format and execution
- Atomicity and transaction ordering
- Compute budget and prioritization

Implementation: Transaction building and submission services

3. Token Standards (SPL Tokens)

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- **Solana Equivalent**: SPL (Solana Program Library) Tokens
- **EVM Comparison**: ERC-20, ERC-721, ERC-1155

Key Topics - Token Standards

- Token Program (SPL-Token)
- Associated Token Accounts (ATAs)
- Token metadata and extensions
- NFT standards (SPL-Token-2022)

Implementation: Token creation, transfer, and management APIs

4. Account Abstraction

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- **Solana Equivalent**: Program-controlled accounts and PDAs
- **EVM Comparison**: EIP-4337 Account Abstraction

Key Topics - Account Abstraction

- Smart accounts via programs
- Programmable transaction authorization
- Session keys and delegated execution

Implementation: Abstracted account management system

5. Fee Mechanism

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- **Solana Equivalent**: Base fees + Priority fees
- **EVM Comparison**: EIP-1559 (base fee + priority fee)

Key Topics - Fee Mechanism

- Fee calculation and prioritization
- Compute unit limits
- Fee markets and congestion handling

Implementation: Fee estimation and transaction optimization

6. Consensus and Validation

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- **Solana Equivalent**: Proof of Stake with Tower BFT
- **EVM Comparison**: Proof of Work (legacy) / Proof of Stake (post-Merge)

Key Topics - Consensus & Validation

- Validator selection and rotation
- Leader schedule and block production
- Fork choice and finality

Implementation: Network monitoring and validation status APIs

7. Signing and Cryptography

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- **Solana Equivalent**: Ed25519 keypairs and signatures
- **EVM Comparison**: ECDSA secp256k1

Key Topics - Signing & Cryptography

- Key generation and management
- Transaction signing workflows
- Multi-signature schemes
- Hardware wallet integration

Implementation: Secure signing services

8. Multi-Party Computation (MPC)

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- **Solana Equivalent**: Threshold signatures and distributed key generation
- **EVM Comparison**: Multi-sig wallets and threshold schemes

Key Topics - MPC

- MPC protocols for secure signing
- Distributed key management
- Threshold cryptography implementations

Implementation: MPC wallet and transaction services

9. Solana Virtual Machine (SVM)

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- **Solana Equivalent**: Sealevel runtime and SVM
- **EVM Comparison**: EVM execution environment

Key Topics - SVM

- Parallel transaction execution
- Runtime architecture
- Program compilation and deployment
- Gas metering and resource limits

SVM Infrastructure

- Local test validator setup
- Network configuration (local/devnet/testnet/mainnet)
- RPC endpoint management
- Faucet integration for development
- Ledger state management
- CLI tools integration

Implementation: SVM integration and program execution APIs

10. Cross-Program Invocations (CPIs)

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- **Solana Equivalent**: Programs calling other programs
- **EVM Comparison**: Contract calls and DELEGATECALL

Key Topics - CPIs

- CPI mechanics and security
- Program composition patterns
- Permission and access control

Implementation: Cross-program interaction services

11. Events and Logging

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- **Solana Equivalent**: Program logs and events
- **EVM Comparison**: Contract events and logs

Key Topics - Events & Logging

- Event emission and parsing
- Log subscription and filtering
- Historical data indexing

Implementation: Event streaming and indexing system

12. Security and Best Practices

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Key Topics - Security

- Common vulnerabilities and mitigations
- Program upgrade patterns
- Access control and authorization
- Audit considerations

Implementation: Security-focused API design

13. Development Tools and Frameworks

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Key Topics - Development Tools

- Anchor framework for Rust programs
- Web3.js and JavaScript tooling
- Testing frameworks and methodologies
- Deployment and monitoring

Implementation: Development tooling integration

14. Network Architecture

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Key Topics - Network Architecture

- Cluster types (mainnet, devnet, testnet)
- RPC nodes and load balancing
- Historical data access
- Network performance optimization

Implementation: Multi-network API support

15. Advanced Features

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Key Topics - Advanced Features

- State compression and accounts database
- Versioned transactions
- Address lookup tables
- Program address introspection

Implementation: Advanced feature demonstrations

Learning Objectives by Module

Beginner Level

- Basic account operations
- Simple token transfers
- Transaction submission
- Network connection and RPC calls

Intermediate Level

- Program development and deployment
- Complex token operations
- Multi-signature transactions
- Event handling and monitoring

Advanced Level

- MPC implementations
- Cross-program invocations
- Performance optimization
- Security hardening

Implementation Roadmap

1. Core account and transaction management
2. Token standards implementation
3. Signing and MPC services
4. Program interaction APIs
5. Event streaming and monitoring
6. Advanced features and optimizations

Design Patterns

Gang of Four (GoF) Patterns

Factory Pattern

- **Application**: Transaction creation and account management
- **Implementation**: Abstract factory for different transaction types
- **Example**: `TransactionFactory.createTransfer()` vs `TransactionFactory.createTokenTransfer()`

Strategy Pattern

- **Application**: Signing mechanisms and fee calculation
- **Implementation**: Pluggable signing strategies (Ed25519, MPC, hardware wallets)
- **Example**: `SigningStrategy` interface with multiple implementations

Observer Pattern

- **Application**: Event monitoring and logging
- **Implementation**: Event listeners for transaction confirmations
- **Example**: `TransactionObserver` subscribing to WebSocket events

Command Pattern

- **Application**: Transaction instructions and program invocations
- **Implementation**: Encapsulated instruction execution
- **Example**: `InstructionCommand` objects for various operations

Adapter Pattern

- **Application**: Multi-chain integrations
- **Implementation**: Unified interface for EVM and Solana operations
- **Example**: `BlockchainAdapter` interface implementations

Blockchain-Specific Patterns

Token Factory Pattern

- **Application**: SPL token creation and management
- **Implementation**: Standardized token deployment with metadata
- **Example**: `TokenFactory.create()` with automatic ATA creation

Multisig Wallet Pattern

- **Application**: Multi-signature transactions and MPC
- **Implementation**: Threshold signature schemes
- **Example**: `MultisigWallet` with configurable thresholds

Oracle Pattern

- **Application**: External data feeds and price feeds
- **Implementation**: Decentralized data sources
- **Example**: `PriceOracle` for DeFi operations

Bridge Pattern

- **Application**: Cross-chain asset transfers
- **Implementation**: Lock-and-mint mechanisms
- **Example**: `BridgeService` for EVM-Solana interoperability

Solana-Specific Patterns

Program Derived Addresses (PDAs)

- **Application**: Deterministic account creation
- **Implementation**: Cryptographic address derivation
- **Example**: `findProgramAddress([userPubkey, "escrow"], programId)`

Associated Token Accounts (ATAs)

- **Application**: Automatic token account management
- **Implementation**: Deterministic token account addresses
- **Example**: `getAssociatedTokenAddress(mint, owner)`

Cross-Program Invocation (CPI)

- **Application**: Program composition and modularity
- **Implementation**: Programs calling other programs
- **Example**: DEX program invoking token program for swaps

Account Abstraction via Programs

- **Application**: Smart accounts and programmable authorization
- **Implementation**: Program-controlled accounts
- **Example**: `SmartAccount` program with session keys

Event-Driven Architecture

- **Application**: Real-time blockchain monitoring
- **Implementation**: Program logs and transaction events
- **Example**: `ProgramEventListener` parsing transaction logs

Resources

- Official Solana Documentation
- Solana Web3.js Guide
- SPL Token Documentation
- Anchor Framework
- Solana Cookbook

Thank You!

Next Steps

- Review TASKS.md for implementation details
- Explore the architecture diagrams
- Start building with the core modules

Happy Learning! 🚀